AIPC Report :

1) What is binning, and what is the trade-off between achievable resolution and signal strength when binning data?

Binning is a technique used in image processing and data analysis where multiple adjacent pixels or data points are combined into larger units called "superpixels" or bins.

The tilt-series was binned using this command: newstack -input TS_33.st -output TS_33_bin2.st -bin 2. We binned by a factor of 2 .After coarse alignment, an additional binning factor of 4 was applied

Binning significantly decreases the data size, leading to faster processing during alignment and averaging steps. We reduce the computation time and we improve Signal-to-Noise Ratio (SNR) by combining adjacent pixels.The primary drawback is the reduction in achievable resolution. Binning by a factor of 2 effectively halves the theoretical maximum resolution.We also lose the Fine Details which affects the accuracy of the final average.

2) What could be a good alternative to patch tracking if the specimen contained gold beads or other spherical high-contrast features?

A good alternative to patch tracking for specimens that have gold beads or other spherical high-contrast features is template matching. This technique uses predefined templates that represent what the high-contrast features are supposed to look like, which helps in accurately locating them within the images. Template matching works particularly well because it takes advantage of the unique shape and contrast of spherical objects, making it easier to identify them even when there is noise in the data. Patch tracking can sometimes struggle with different structures, while template matching tends to be more accurate and reliable when working with spherical particles.

3) How did you optimize alignment parameters to reduce the residual error mean?

To optimize the alignment parameters and reduce the residual error mean in our tomogram reconstruction, I adjusted several settings. I set the **size of the patches** to **800 x 800 pixels** . This provided a good balance between capturing sufficient detail and minimizing computational load. A **fractional overlap** of **0.33** was implemented to ensure that adjacent patches shared enough information, facilitating smoother transitions and improving alignment accuracy. I increased the **number of iterations** to improve subpixel accuracy, which contributed to reducing the residual error mean significantly, even if it did not reach the target threshold .Additionally, I specified a **tomogram thickness of 300 nm** to encompass all relevant features while limiting empty space, which is crucial for accurate averaging. The target residual error for the alignment was set at 0.35 nm, and while I did not fully achieve this goal, the results were still satisfactory for our analysis.

4) Speculate why allowing more than one "spike" might be problematic considering their irregular distribution on the virus surface.

Allowing more than one spike in the analysis of the virus surface can be problematic due to the irregular distribution of these spikes, which may lead to inaccuracies in subtomogram averaging. The spikes, which represent the viral envelope proteins, do not have a uniform arrangement on the virus surface; instead, they are often unevenly spaced and may vary in orientation. If multiple spikes are included in a single subtomogram, the averaging process could incorporate conflicting spatial information from these irregularly positioned features. This could result in a blurred or distorted average that does not accurately reflect the true structure of the spikes, ultimately compromising the resolution and reliability of the reconstructed model. Furthermore, the presence of multiple spikes may introduce additional noise and variability into the dataset, making it challenging to achieve a clear and precise representation of each individual spike's conformation and arrangement. Therefore, it is crucial to isolate individual spikes during analysis to ensure accurate structural determination and interpretation.

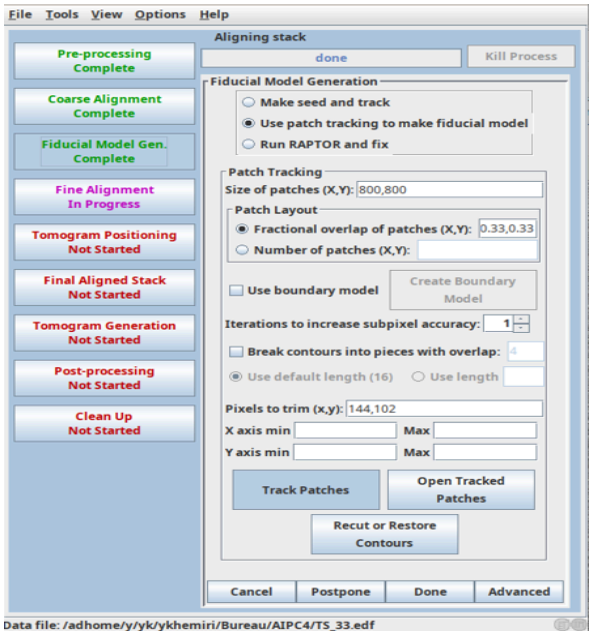
I applied a cylindrical mask with a radius of 20 units and a height of 35 units and to achieve proper alignment, I implemented a 10-unit shift to center the mask accurately over each individual spike .

5) why and how this allows you to reduce the azymuth rotation range. Also explain how you have chosen the cone aperture.

Reducing the azimuth rotation range is essential in subtomogram averaging, especially when dealing with symmetrical structures like the ENV protein, which is presumed to form a trimer with c3 symmetry. By recognizing this symmetry, we can limit the azimuthal rotation range to one-third of a full rotation (120°), effectively reducing the computational burden and enhancing the averaging process. This approach allows us to capture the essential orientations of the spikes while minimizing redundancy in the data. Furthermore, by focusing on a smaller range of angles, we can improve the signal-to-noise ratio in our averages, as fewer orientations can lead to cleaner averages without losing critical structural information.

When selecting the cone aperture for our analysis, I chose a cone aperture of 60 degrees to effectively balance the need for angular diversity with the goal of minimizing background noise and irrelevant features. This aperture size allows us to capture a sufficient range of orientations around the spikes while ensuring that we focus primarily on the relevant structural details of interest. By using a 60-degree cone aperture, we can adequately sample the orientations of the ENV protein spikes, which is crucial for accurate subtomogram averaging. This selection helps us optimize the averaging process by influencing the overall quality and fidelity of our final structural model.

Annexe :



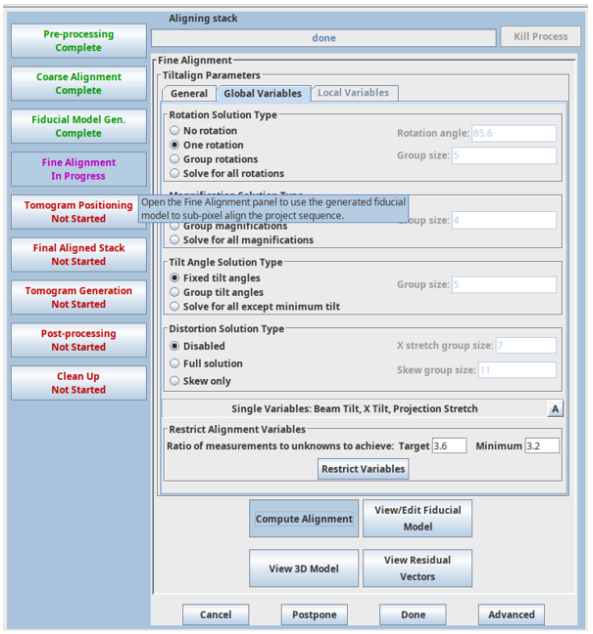


Fig 1a: Project log Left : the alignment parameters : Patch Size, Fractional Overlap and iterations **b.** Another project log

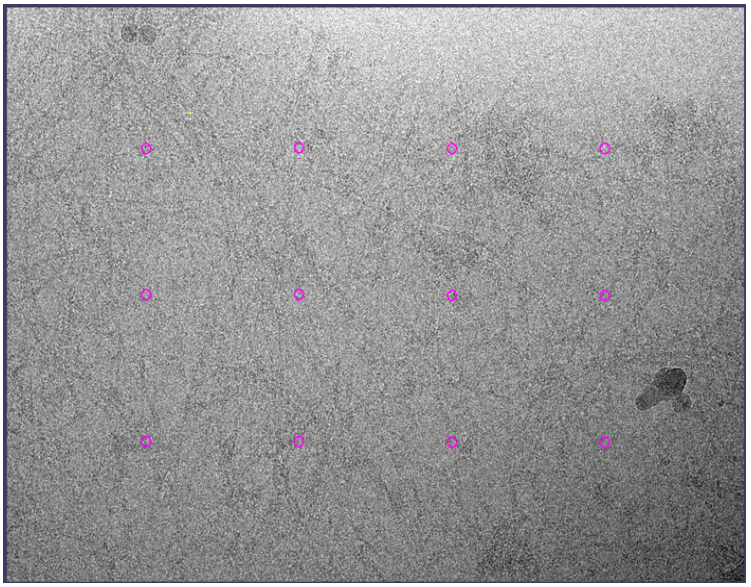


Fig 2: This figure shows the tilt-series view with fiducial markers in magenta used for alignment.

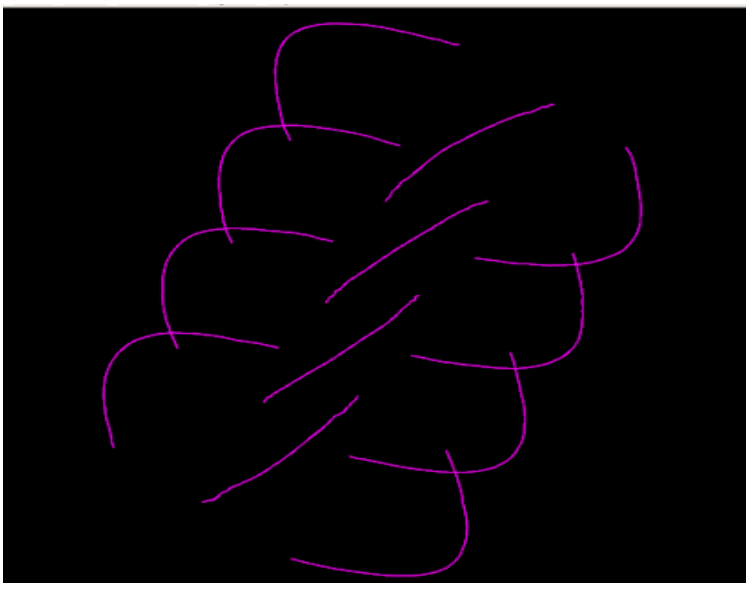
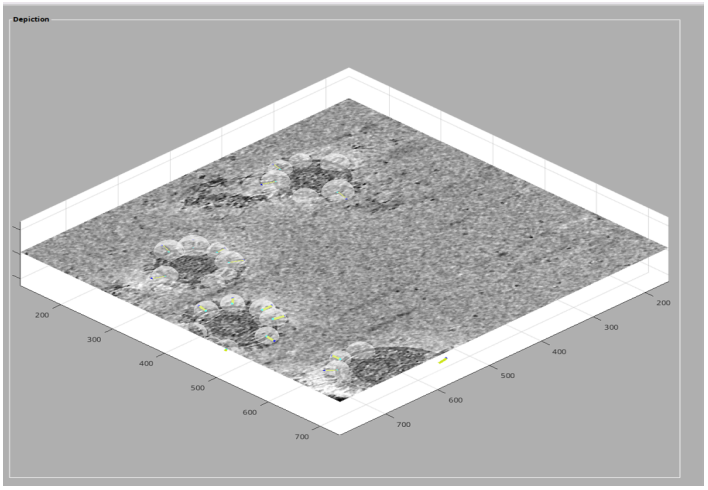


Fig 3 : It shows the corresponding 3dmod model view of the fiducial positions .

Fig 4: The fully reconstructed tomogram generated from the aligned tilt-series showing filamentous structures



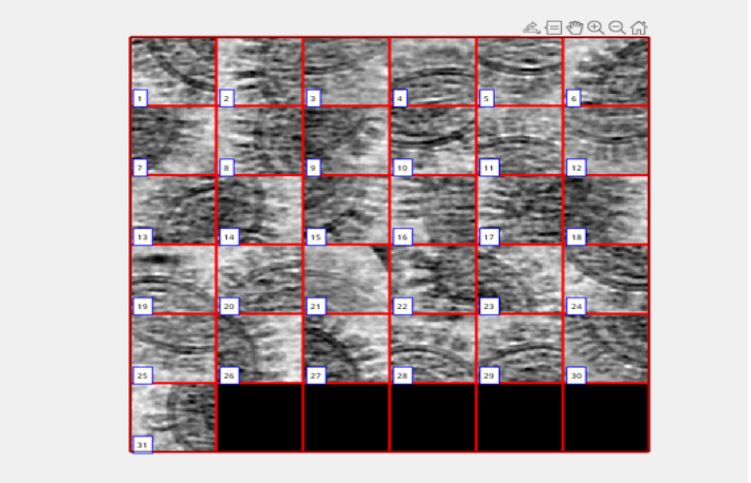


Fig 5 : a. Left : This figure shows our Particle Picking step **b. Right panel :** It displays The picked particles

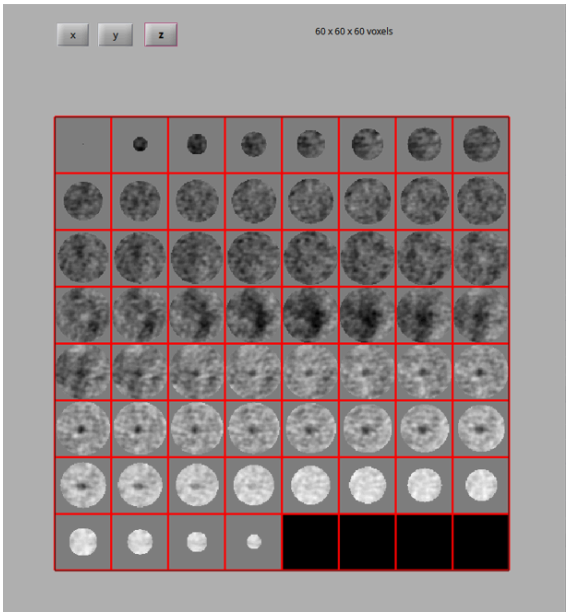


Fig 6: This is a display of our masked initial averages in Z axis

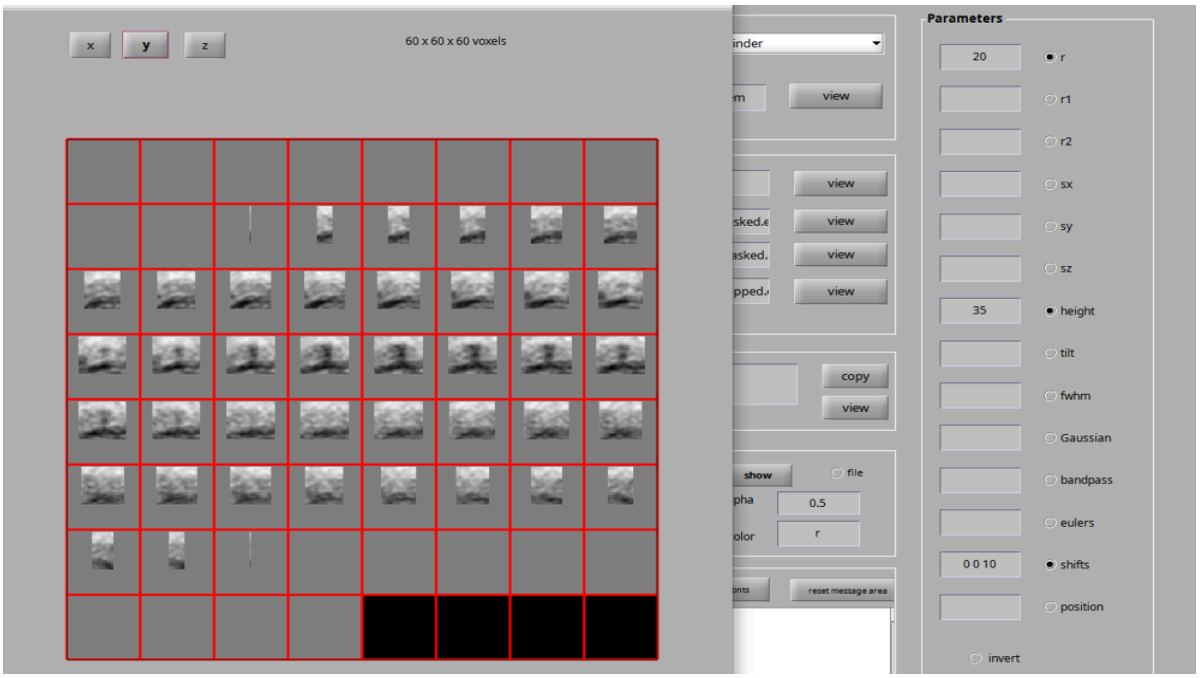
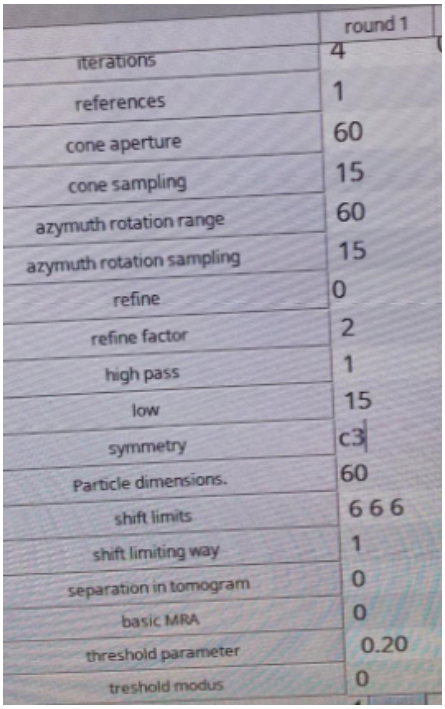


Figure 7 : this figure shows our masked initial averages in y axis and my mask parameters



	round 1
iterations	4
references	1
cone aperture	60
cone sampling	15
azymuth rotation range	60
azymuth rotation sampling	15
refine	0
refine factor	2
high pass	1
low	15
symmetry	c3
Particle dimensions.	60
shift limits	6 6 6
shift limiting way	1
separation in tomogram	0
basic MRA	0
threshold parameter	0.20
treshold modus	0

Fig 6 : A screenshot of the numerical parameters

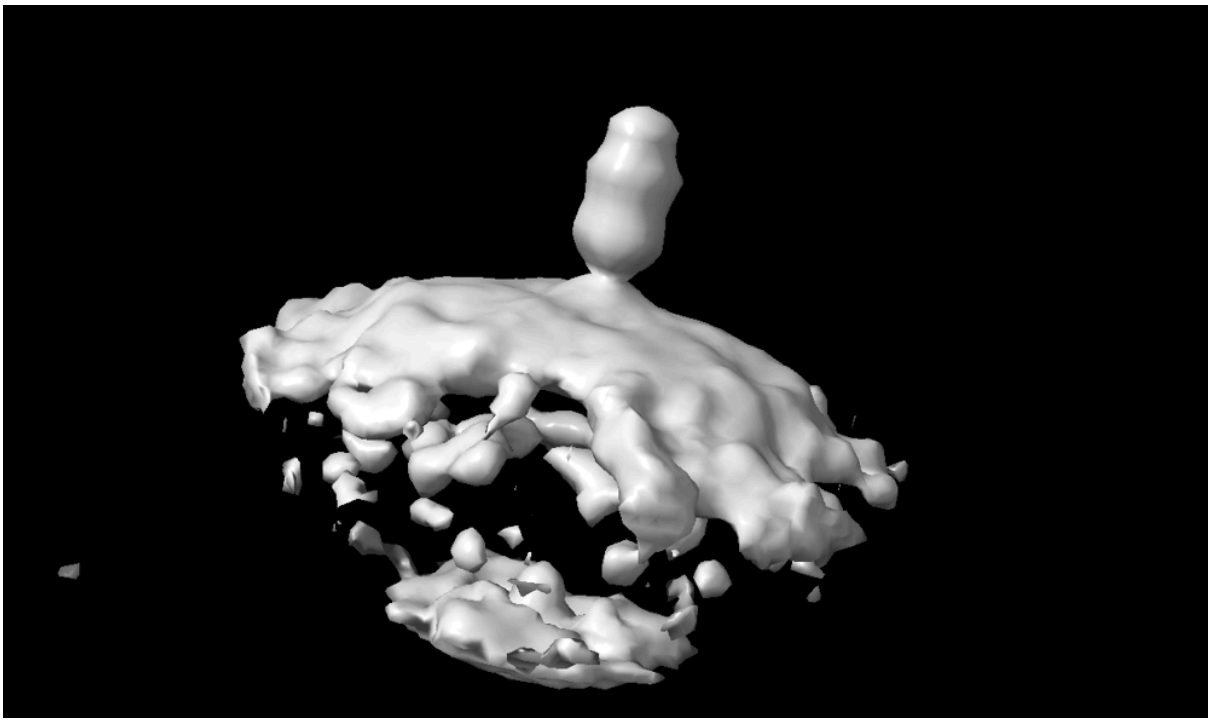


Fig 7 : The final average of my model visualized in Chimera using this command : volume capFaces False.